Detection of Common Allergens Using Escherichia coli and Ligand-Mediated Aptamer-Ribozyme Complexes

Abstract

Many food products contain allergens without proper precautionary labelling, leaving consumers at risk of an unexpected reaction. Current food test kits on the market are expensive, specific to one allergen, and may take days to process within a lab setting. This work proposes using aptazymes within a synthetic biological circuit in BL21(DE3) *Escherichia coli* (*E. coli*) to detect and report the presence of casein, galactose, and gliadin, improving cohesion and accessibility to allergen detection systems. MATLAB SimBiology modelling of the system using 500 µM of initial allergens present in the cell demonstrated detectable peak expression of 10µM amilCP for the gliadin detection system, while the AND gate system that simultaneously detects casein and galactose was unable to express eforRed at detectable levels, peaking at 5e-46M. Future work could include computational model optimization to improve accuracy, and in vitro validation to determine expression time and system sensitivity.

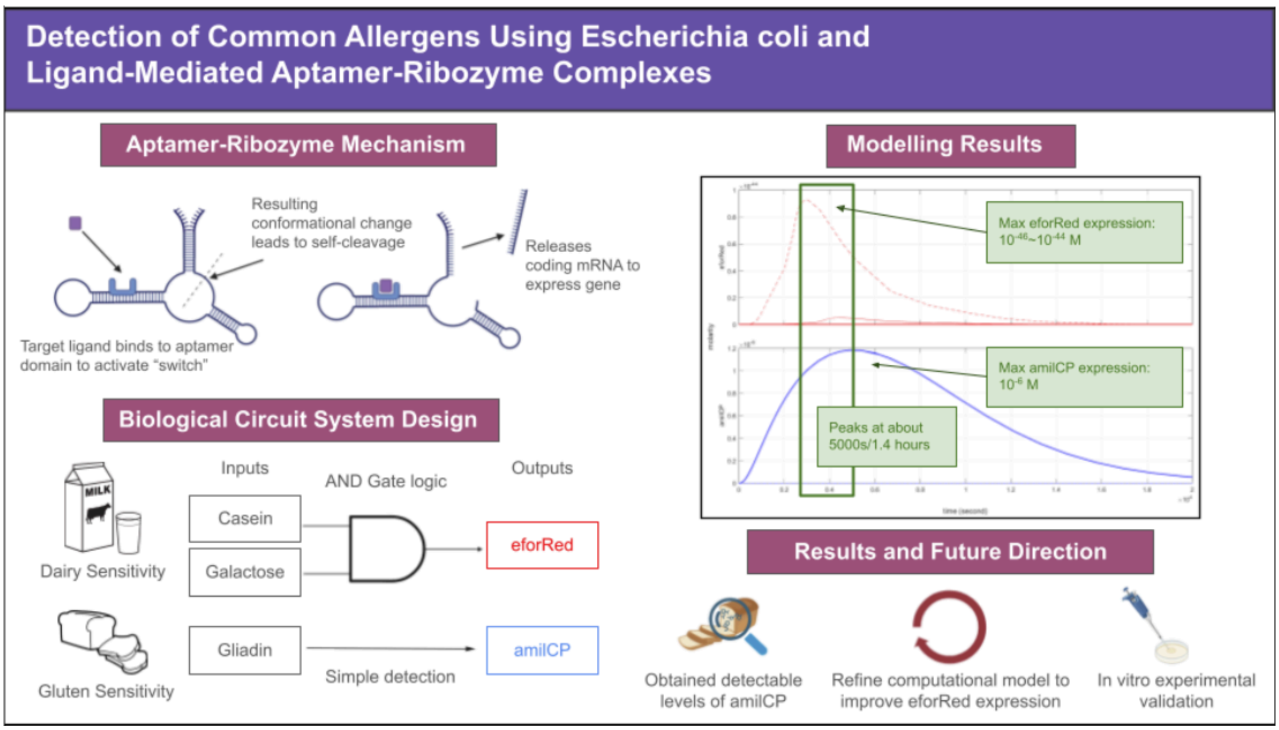

Figure 1. Graphical abstract summarizing the work done on this project, including a concise summary of the aptazyme mechanism (top left), a simplified overview of the biological circuit and target allergens (bottom left), summary of key model outputs (top right), and a graphical summary of results and future directions (bottom right).

Introduction

Allergen exposure is a critical global health issue and can trigger a cascade of symptoms ranging from nausea and vomiting to life threatening bronchospasm and syncope [1]. In Canada, Approximately 9.3% of Canadians of all ages reported a perceived food allergy in 2016 [2]. As a result, food allergies impose a constant psychological and social burden, requiring individuals to remain vigilant about eating.

A primary difficulty in managing food allergies is the prevalence of unintended allergen exposure, particularly through cross-contamination at various stages of food processing, handling, and manufacturing [3]. A prospective cohort study performed by Blom et al. 2018 found that around 20% of 51 food product samples contained detectable allergens without any precautionary labelling [4].

Currently, consumers rely on preventative, avoidance, or reactive emergency medicine [5]. Existing detection methods such as ELISA or PCR-based testing are the gold standard for accuracy, but suffer from prohibitive constraints. These methods are expensive, require specialized laboratory equipment and lengthy lab processing times [6]. Currently, there is no accessible, low-cost method for a consumer to verify the safety of their food.

Synthetic biology has been proposed as a solution to this diagnostic gap by repurposing micro-organisms into "living" sensors to detect environmental contaminants, such as mercury and lead [7]. However, the practical deployment of these cell-based biosensors is hindered by significant technical hurdles, including the need for long-term shelf stability and the persistent risk of false positives, a challenge often seen in traditional assays like ELISA, where prioritizing high sensitivity frequently comes at the expense of specificity [8], [9].

Using BL21(DE3) *E. coli* as a bacterial chassis, these concepts can be further refined through the integration of aptazymes; molecular switches engineered to bind specific ligands [10]. An aptazyme consists of two parts; a functional single-stranded RNA (ssRNA) aptamer that binds to target ligands, and a hammerhead ribozyme (HHR), another functional ssRNA that self-cleaves upon site-specific hydrolysis, releasing a mRNA strand coding strand. Upon binding a target allergen, an aptazyme self-cleaves to release a sequestered mRNA, which is then translated into an intermediate messenger [11]. This messenger activates specific promoters to drive the production of chromoproteins, resulting in a clear, color-coded visual readout.

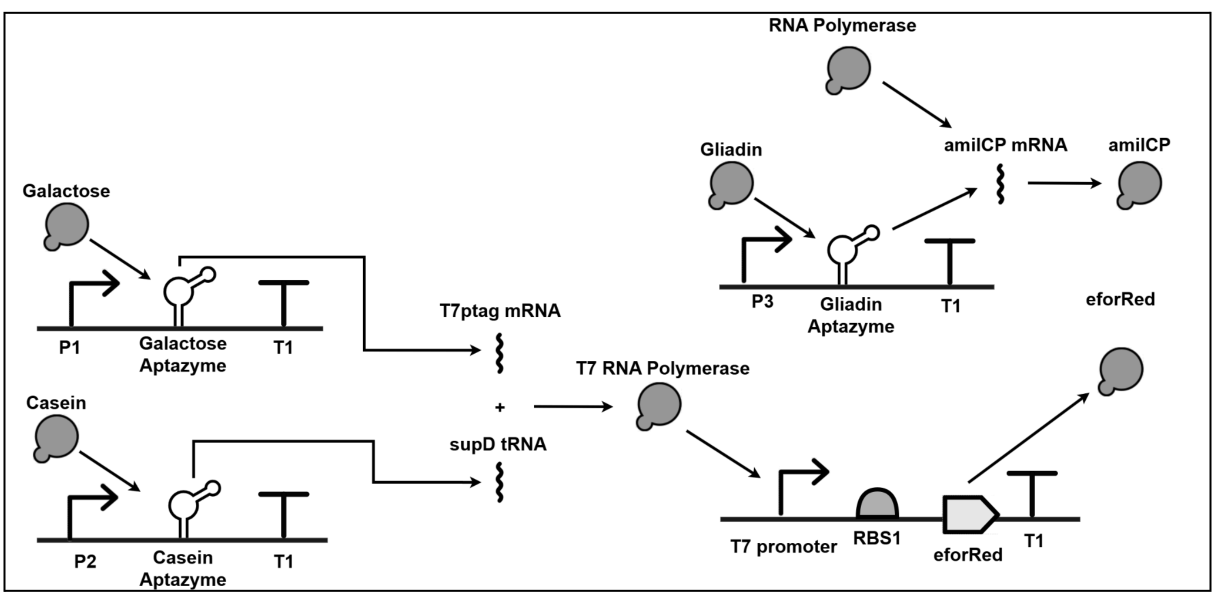

Figure 2. SBOL diagram of the proposed biological system in. All aptazymes are constitutively produced inside the bacteria with constitutive promoters P1 (BBa_J23100), P2 (BBa_J23101), and P3 (BBa_J23102)

for the casein, galactose, and gliadin aptazymes respectively. Aptazymes are used to detect each allergen and release mRNA strands that translate into intermediate transcriptional regulators. The galactose aptazyme releases SupD tRNA, which allows functional translation of T7ptag mRNA released by the casein aptazyme through amber stop codons, represented by '+'. The casein and galactose detection system utilises the T7 promoter system to form an AND gate that expresses the red chromoprotein eforRed, and the gliadin detection is a parallel circuit that expresses blue chromoprotein amilCP.

Materials and Methods

The biological system will be composed of the pSB1C3-pT7-eforRed and pBBR1MCS2-Casein-T7ptag-Galactose-supD-Gliadin-amilCP plasmids, transformed into BL21(DE3) *E. coli* supplied by Goldbio [12]. All reagents and proposed biological parts used are available in Table 4 and Tables 1-3 respectively in Appendix A.

To ensure the system's efficacy, the plasmid responsible for the constitutive production of aptazymes will be constructed via Gibson assembly. This method was selected for its high fidelity, which is critical given that the functionality of the aptazyme system is dependent on the precise folding of the resulting RNA sequences [13], [14]. Due to the complexity of the aptazyme sequences, each RBS-Aptamer-HHR-CDS insert would need to be ordered as a single unit, with design rationale discussed in Appendix B [13], [15], [16], [17], [18], [19], [20]. Overhangs that produce complementary sequences following exonuclease digestion will be attached to parts using PCR [21]. Following assembly, a portion of the reaction product will be digested with PstI and BsaI to produce easily identifiable fragments (5.3kbp, 2.7kbp, 1.2kbp, 0.3kbp) and will be run through gel electrophoresis to ensure that the assembled product's size and identity match the expected result [22]. The final plasmid is shown in Figure 3.

The AND gate logic plasmid will be constructed using BioBrick assembly to take advantage of the well-characterized parts, simple testing methodology and verification [23]. This method utilizes standardized flanking sequences (Prefix and Suffix) containing specific restriction sites. When two parts are ligated, they form a "scar" that maintains the frame while preserving the original flanking sites for subsequent insertions [24]. BioBrick assembly will be performed twice, and each product will be verified with gel electrophoresis following EcoRI-HF and PstI digestion to catch potential errors. After attaching BioBrick prefixes and suffixes to each insert using PCR, the pSB1C3 backbone will be opened using EcoRI-HF and PstI and have the pT7-RBS sequence (digested using EcoRI-HF and SpeI) inserted upstream of the eforRed CDS (digested with XbaI and PstI). The second assembly will be a suffix insertion of the double terminator, opening the plasmid with SpeI and PstI and digesting the insert with XbaI and PstI. The final plasmid is shown in Figure 4.

Following successful assembly, both plasmids will be transformed into BL21(DE3) *E. coli* using electroporation. This method was selected for its high transformation efficiency, important because two plasmids are being transformed into the chassis [25]. Transformation will be verified by plating cells on agar plates containing both kanamycin and chloramphenicol. Since each plasmid backbone contains one of kanamycin or chloramphenicol resistance, surviving colonies would have been successfully transformed with both plasmids [26].

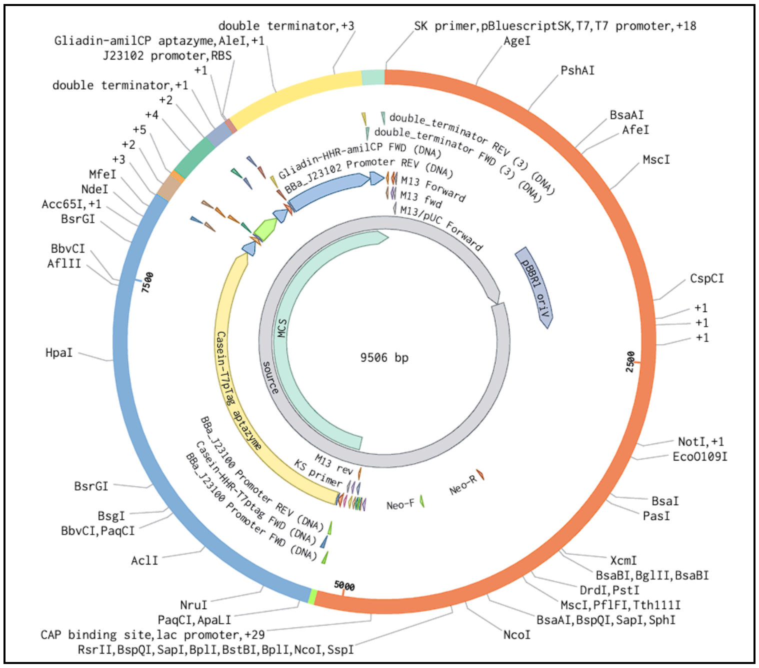

Figure 3. Plasmid map of pBBR1MCS2-Casein-T7ptag-Galactose-supD-Gliadin-amilCP to be assembled using Gibson Assembly. The pBBR1MCS-2 backbone contains a kanamycin resistance gene used to select successful transformants. The primers would be attached to each insert using PCR, and pBBR1MCS2-Casein-T7ptag-Galactose-supD-Gliadin-amilCP would be assembled in one reaction. Each RBS-aptamer-HHR-CDS insert would need to be ordered as a single unit, with design rationale discussed in Appendix B.

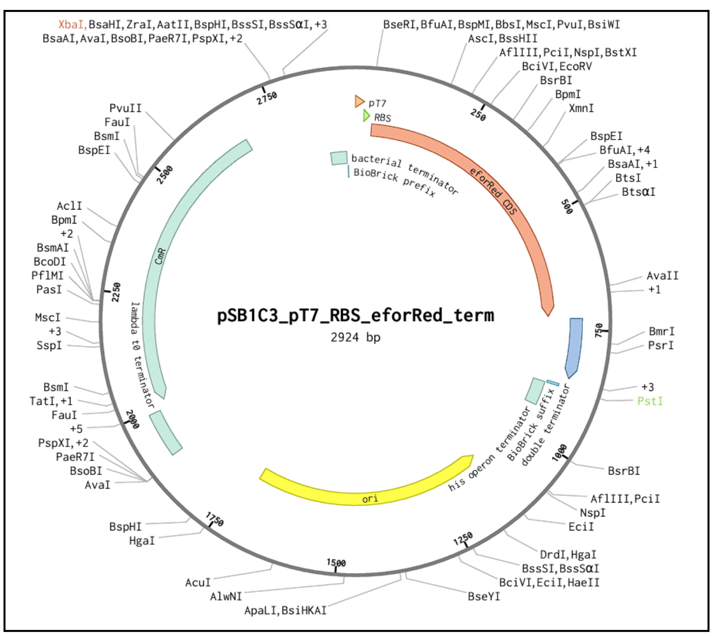

Figure 4. Plasmid map of pSB1C3-pT7-eforRed to be assembled using BioBrick Assembly. The pSB1C3 backbone contains a chloramphenicol resistance gene used to select successful transformants, and Biobrick prefix and suffix to facilitate BioBrick Assembly. The BioBrick prefix and suffix would be attached to each

insert using PCR, and pSB1C3-pT7-eforRed would be assembled in two reactions. The first reaction would insert pT7-RBS upstream of the eforRed CDS in the pSB1C3 backbone, and the second reaction would be a suffix insertion of the double terminator downstream of the eforRed CDS.
While in vitro experiments are outside the scope of this project, a computational model was created and analyzed to serve as in silico validation. The proposed system was constructed using a dynamic ODE based model, which are commonly used to model complex biological systems [27]. The model simulated the transcription, binding, and self-cleavage of the aptazymes with their respective ligands and mRNA products, the translation of mRNA into their respective proteins, and the degradation of all mRNA, protein, and complex species, with constants and rates shown in Tables 1-3. The model uses constants for a system temperature of 37°C, as it is the optimal temperature for cultivating *E. coli* [28]. An SBOL diagram summarizing the major reactions of the system can be seen in Figure 2. The model simulated the transcription of aptazymes under the constant rate assumption, degradation of all species under the zero rate assumption, and the translation of the chromoproteins, the transcription of the eforRed mRNA, and degradation of all species under the mass action rate assumption.

The degradation of the aptamer-ligand complexes was assumed equal to the degradation of mRNA, as mRNA degrades faster than the ligands they bind to [29]. It was assumed that the ligands had an initial concentration inside the cell of 0M, 500µM, or 1M depending on the test conditions [30], and all other species had an initial concentration of 0M inside the cell. Aptazymes were assumed to not spontaneously cleave without binding to a ligand, under the zero rate assumption [31], and all aptazymes were assumed to have the dissociation and association constants of the Lac 20 aptamer [32]. The model assumed that all proteins have the same loss rate, all mRNA had the same loss rate, and that aptazymes only bind to the ligands that they were intended to detect [33]. Galactose was assumed to have the same degradation rate as glucose [34].

Results and Discussion

Several Simbiology simulations were performed with the presence and absence of various combinations of ligands as outlined in the experimental validation. A ‘present’ signal was modelled with the initial ligand concentration set to 500µM, with justification supplied in Appendix B, whereas ‘absent’ was modelled with an initial ligand concentration of 0M, as they aren’t natively produced. Figure 5 displays the simulation results of all relevant combinations of ligands. As seen in Figure 5, eforRed and amilCP concentrations both reached their peak after around 5000 seconds (1.4 hours). eforRed had a maximum concentration of about 5e-46M, and amilCP of about 1.2µM. To determine the effects of casein and galactose on eforRed expression, test simulations with these ligands at initial concentrations of 1M were also performed, seen in Figure 6. When casein concentration was raised to 1M, the maximum eforRed concentration increased to approximately 9e-44M. However, when galactose concentration was raised to 1M, eforRed concentration increased dramatically, peaking at approximately 5e-34M. When both casein and galactose concentrations were raised to 1M, the peak concentration of eforRed was still approximately 5e-34M, as seen in Figure 6. Evidently, increasing the concentration of galactose has a much greater impact on the final level of eforRed.

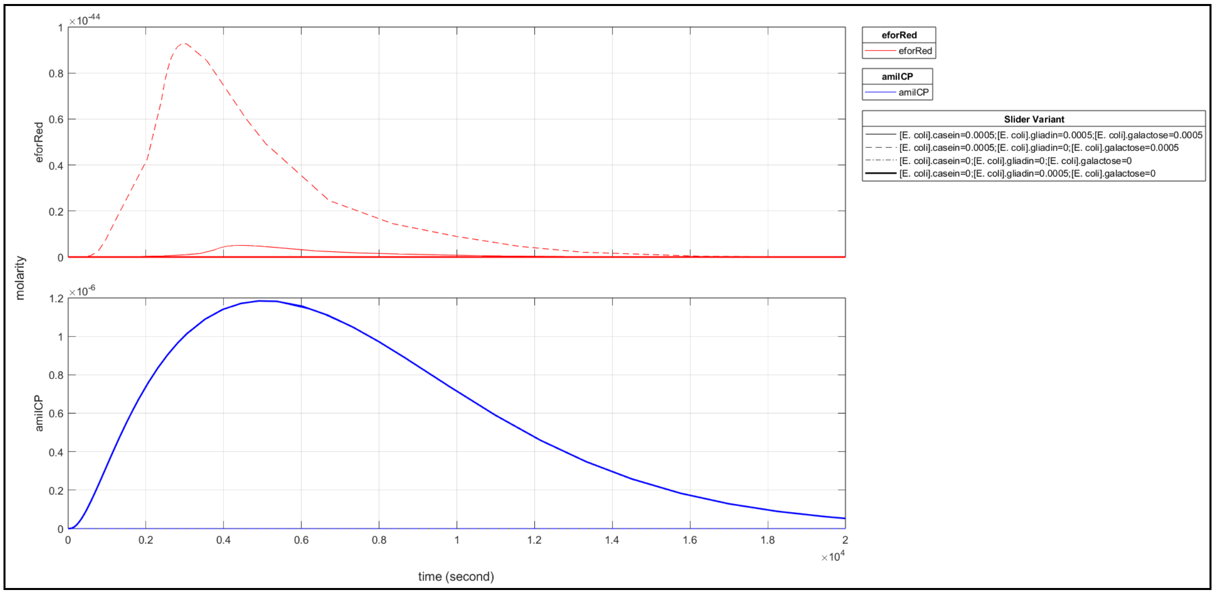

Figure 5. eforRed (red, top) and amilCP (blue, bottom) molarity over time shown in red, plotted under four different conditions: 500µM casein, 500µM galactose, and 500µM gliadin ( ), 500µM casein, 500µM galactose, and 0M gliadin ( ), 0M casein, 0M galactose, and 0M gliadin ( ), and 0M casein, 0M galactose , and 500µM gliadin ( ). All other proposed test scenarios appear as expected and have been omitted for clarity, but can be found in Figure 7 in Appendix C.

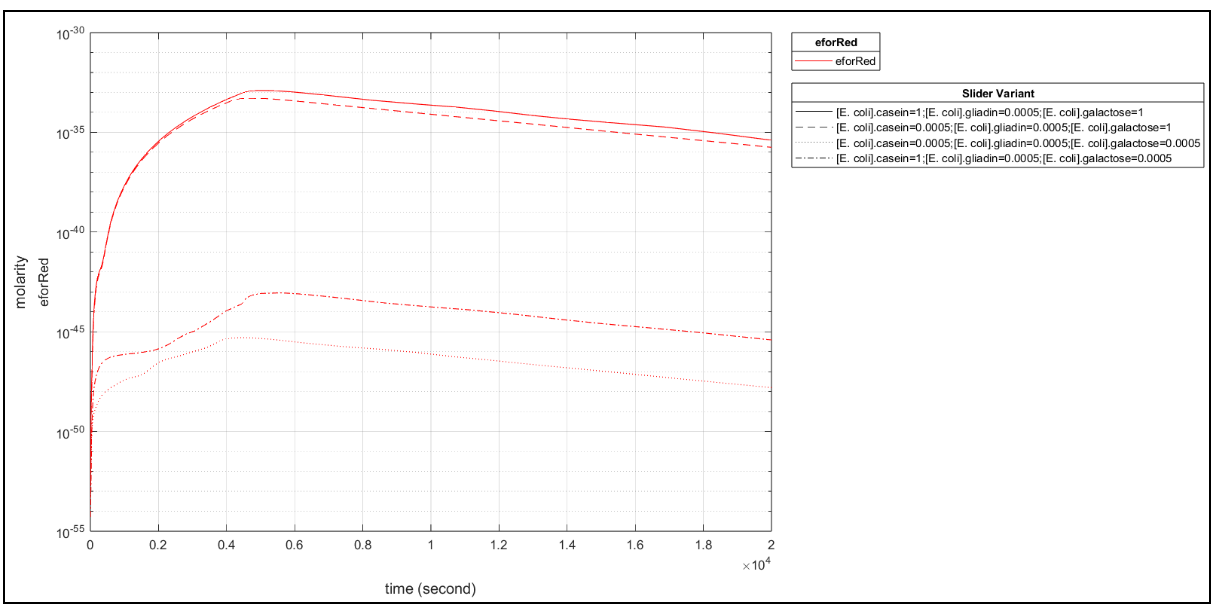

Figure 6. eforRed molarity over time shown in red, plotted on a semilog-y plot under four different conditions: 1M casein and 1M galactose ( ), 500µM casein and 1M galactose ( ), 500µM

casein and 500µM galactose ( ), and 1M casein and 500 µM galactose ( ). Initial gliadin concentration was 500µM throughout.

Both amilCP and eforRed were produced, reached a peak concentration, and eventually degraded, which was expected behavior for the model. A major discrepancy in expected behaviour was the low level of eforRed expression, especially in comparison to amilCP expression. This was likely due to the number of intermediates involved in eforRed expression, since each intermediate would degrade slightly before being able to transform into eforRed, leaving the final concentration orders of magnitude lower than the initial concentration of ligands.

Using the Beer-Lampert law outlined in Appendix B, the minimum detectable concentration of amilCP and eforRed are found as 0.11µM and 0.32µM respectively. The predicted peak amilCP concentration of 1.2µM is above the detection threshold, but the eforRed concentrations are significantly below the detection threshold, even under extreme conditions.

The current model uses high tolerances to ensure computational functionality within MATLAB SimBiology. A more accurate model of the system could better inform which aspects of the genetic circuit need to be improved to achieve the desired output. Should that model continue to produce low eforRed concentrations, additional modifications to the logical AND gate system will need to be made to improve sensitivity and mitigate intermediate degradation, potentially informed by in vitro testing. The modelled expression of both chromoproteins peak at around 5000 seconds, or 1.4 hours after transportation into the cell, as seen in Figures 5-8. This is significantly faster than current solutions [35]. This relatively fast expression determined in silico could inform the time intervals at which chromoprotein expression is measured in vitro.

A key assumption limiting accuracy is that the target ligands begin in the cell. In reality, these ligands must be uptaken into the cell through specific membrane transport mechanisms, a process governed by uptake rate constants currently omitted from the model [36]. Neglecting these transport kinetics overestimates the response speed and sensitivity of the biosensor.

The model also assumes that aptazymes self-cleave only when bound to the target ligand. This ignores the possibility of premature self-cleavage or incomplete ribozyme activation. By failing to account for aptazyme fouling and non-specific interactions, the model likely overestimates sensitivity and fails to capture the background noise inherent in real-world biological systems [31].

The model also assumes a lack of interference between the two parallel circuits and treats them as perfect orthogonal systems. In practice, the systems would compete for a finite pool of shared resources, such as RNA polymerases and ribosomes. This model also uses generalized transcription, translation, and degradation rates, not accounting for differing rates between species. The competition between the circuits combined with the lack of specific kinetic rates introduce inaccuracies that skew the predicted steady-state expression levels.

Following successful in silico validation, the system will be tested in vitro to determine sensitivity and speed of expression. The transformed *E. coli* should be tested under every combination of ligands, present or absent, at varying concentrations. The plate with no added ligands will serve as a negative control, and a culture with only added gliadin can act as a positive control for the aptazyme, since producing amilCP will confirm aptazyme binding can produce a product. For the logical AND gate, the bacteria should produce eforRed in the presence of both casein and galactose, and nothing in the absence of either of them. Testing with all three ligands is also necessary to ensure the systems can function together without interference and produce both chromoproteins. The expected experimental results can be seen in Table 8 in Appendix A.
As the circuit produces chromoproteins, it can be verified to function by measuring the absorbance of the cell plate. eforRed has an absorbance peak at 224nm [37], and amilCP at 588nm [38], so using a spectrophotometer to measure absorbances of the cell suspension at these wavelengths can determine the concentrations of the chromoprotein products. While amilCP has a similar absorbance peak to *E. coli* at 600nm, previous research has shown that it is still detectable in bacterial culture [38].

Other factors to consider include the pH and ion concentration of the medium, as these factors impact hammerhead ribozyme activity, Mg^{2+} being particularly important for HHR function [39]. Thus, it would be ideal to test the system in various pH and concentrations of Mg^{2+} to determine optimal conditions for aptazyme functionality.

If the system works as expected, then absorbance measurements can be directly correlated to chromoprotein and ligand concentrations. If the absorbance measurements are below what is expected given the concentration of ligand added, it must be determined at what point(s) the system failed. Unsuccessful ribozyme cleavage can be detected by lysing the cell and performing gel electrophoresis to determine if the expected bands are absent. If the mRNA was not successfully cleaved, it may be the *E. coli* failing to uptake ligands, or the aptamers failing to recognize the ligands. The conformational change of the aptazyme binding may be detected using microwaves, measuring the electromagnetic properties of the aptazyme to indicate successful ligand binding [40]. Cellular uptake of ligands may be confirmed using fluorescent labelling of ligands, and looking for fluorescence inside the cell [41].

Conclusion

The results of this study demonstrate the feasibility of using ligand-mediated aptazyme circuits for allergen detection, though with varying degrees of success across subsystems. The MATLAB SimBiology model revealed that the gliadin sensing system successfully produced amilCP at a peak concentration of approximately 1.2 μM, which is above the established detection threshold of 0.11 μM required for a reliable readout. However, the AND gate system designed to detect casein and galactose failed to reach detectable levels of eforRed. Even under extreme conditions with 1M ligand concentrations, the eforRed expression remained several orders of magnitude below the 0.32 μM detection limit, likely due to the high number of molecular intermediates and the degradation of those species before eforRed translation could occur.

The primary limitations of this study stem from modeling assumptions and data availability; specifically, the inability to source precise dissociation values for the aptazymes. Consequently, the model relied on the dissociation and association constants of the Lac 20 aptamer for all aptazyme interactions, which does not accurately reflect the unique binding affinities of casein, galactose, or gliadin. Future research should prioritize the empirical determination of these kinetic constants to improve model accuracy. Optimizing the AND gate architecture to reduce intermediates, increasing the stability of the mRNA transcripts to ensure that reporters like eforRed can accumulate to detectable levels, and in vitro validation to characterise expression time and system sensitivity are also potential areas of improvement.

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Appendix A - Tables

Table 1. Aptamer and ribozyme sequences used for the construction of pBBR1MCS2-Casein-T7ptag-Galactose-supD-Gliadin-amilCP.

Type	Name	Function	Supplier	Reference
Hammerhead ribozyme	HHR	Self cleavage upon activation to release mRNA	Twist Bioscience	[42]
Aptamer	N/A	Detects and binds to gliadin	Antiboides.com limited	[43]
Aptamer	N/A	Detects and binds to casein	Amsbio LLC	[44]
Aptamer	Galactose-bisboronic acid aptamer-sensor	Detects and binds to galactose	Aptagen LLC	[45]

Table 2. External primer sequences used for the construction of pSB1C3-pT7-eforRed, all other primer sequences were generated in Benchling.

Sequence Name	Sequence	Supplier	Description and Reference
BioBrick Prefix	gaattcgcggccgcttcta	iGEM Registry	Prefix for all forward primers below that are used in BioBrick [46]
BioBrick Suffix	ctgcagcggccgctactagta	iGEM Registry	Suffix for all forward primers below that are used in BioBrick [46]

Table 3. Host bacteria, plasmid backbones and other biological part sequences for the construction of pBBR1MCS2-Casein-T7ptag-Galactose-supD-Gliadin-amilCP and pSB1C3-pT7-eforRed

Type	Name	Function	Supplier	Reference
Host Cell	BL21(DE3) Electrocompetent *E. coli* Cells	Contain recombinant plasmids for sensor system	GoldBio	[47]
Plasmid Backbone	pBBR1MCS-2	Low-copy plasmid with kanamycin resistance	Addgene	[48]
Plasmid Backbone	pSB1C3	High copy plasmid with chloramphenicol resistance	Addgene Inc.	[49]
Chromoprotein	eforRed	Chromoprotein reporter (red), expressed in presence of galactose and casein	iGEM Registry	[37]
Chromoprotein	amilCP	Chromoprotein reporter (blue), expressed in presence of gliadin	iGEM Registry	[38]
CDS	T7ptag Polymerase (with amber mutation)	Function as inactive element to be activated in AND gate	iGEM Registry	[50]
tRNA	SupD-tRNA	Suppresses amber codon in T7ptag to allow it to activate AND gate of T7 promo	iGEM Registry	[51]
Promoter + RBS	Promoter T7 and RBS	T7 promoter with RBS	iGEM Registry	[47]
Double	BBa_B0015	Double terminator	iGEM Registry	[52]
Constitutive Promoter	BBa_J23100	Anderson collection constitutive promoter	iGEM Registry	[53]
Constitutive Promoter	BBa_J23101	Anderson collection constitutive promoter	iGEM Registry	[54]
Constitutive Promoter	BBa_J23102	Anderson collection constitutive promoter	iGEM Registry	[55]
RBS	BBa_B0030	Strong RBS	iGEM Registry	[56]

Table 4. Other materials and reagents for experiments and protocols.

Name	Function	Supplier	Reference
BioBrick Assembly Kit	To perform RE digests for Golden Gate Assembly (includes restriction enzymes, buffers, loading dyes)	New England Biolabs	[57]
GeneArt™ Gibson Assembly HiFi Master Mix	To perform Gibson Assembly	Thermofisher Scientific	[58]
Monarch Spin Plasmid Miniprep Kit	*E. coli* miniprep kit	New England Biolabs	[59]
Certified Molecular Biology Agarose	Agarose powder for casting gels for electrophoresis	Bio-Rad	[60]
50x Tris/Acetic Acid/EDTA (TAE)	TAE buffer for casting gels and running gel electrophoresis	Bio-Rad	[61]
PCR Mastermix	Contains Taq Polymerase, dNTPs, and buffers for PCR reaction	Thermofisher Scientific	[62]
Rapid DNA Ligation Kit	Contains T4 DNA ligase and 5x reaction buffer	Thermofisher Scientific	[63]
ddH_2O	Electroporation reagent	Sigma	[64]
Glycerol	Electroporation reagent	Sigma	[65]
Super Broth	To grow the bacteria in	BioShop Canada	[66]
Kanamycin	Antibiotic for selection	Sigma	[67]
Chloramphenicol	Antibiotic for selection	Sigma	[68]
Agar powder	To pour agar plates	Sigma	[69]
Gliadin	To test detection system	Sigma	[70]
Casein	To test detection system	Sigma	[71]
Galactose	To test detection system	Sigma	[72]

Table 5. Table of parameters and values used in outlined mathematical model

Name	Value	Unit	Description	Reference
Cell Volume	4.4	fL	Volume of 1 Cell of *E. coli*	[73]
k_on	96	$M^{-1}sec^{-1}$	Aptazyme association constant	[31]
k_off	8.30e-02	sec^{-1}	Aptazyme dissociation constant	[31]
k_cleave	0.069	sec^{-1}	Aptazyme cleavage constant	[31]
k_pro_loss	3.85e-04	sec^{-1}	Protein degradation rate	[33]
k_mRNA_loss	2.31e-03	sec^{-1}	mRNA degradation rate	[33]
k_trsc	1.2e-10	$M*sec^{-1}$	mRNA transcription rate	[33]
k_trsl	0.01527	sec^{-1}	Protein translation rate	[33]
SupD_loss	6.667e-02	sec^{-1}	SupD degradation rate	[74]
gal_loss	0.4167	sec^{-1}	Galactose degradation rate	[34]
T7ptag_loss	6.667e-02	sec^{-1}	T7ptag degradation rate	[74]
T7_loss	5.833e-02	sec^{-1}	T7 RNA polymerase degradation rate	[75]
k_trsl_T7	0.04445	sec^{-1}	T7 RNA polymerase translation rate	[75]
k_trsc_T7	0.0587 (40bp/sec/681bp)	sec^{-1}	T7 polymerase transcription rate	[76]
K_T7	2.000e-09	M	T7 dissociation constant	[74]

Table 6. Table of species used in the outlined mathematical model, all measured in units of molarity

Name	Initial Conditions	Description	Reference
casein	0M/5e-4M/1M	Casein concentration inside the cell	[77]
galactose	0M/5e-4M/1M	Galactose concentration inside the cell	[30]
gliadin	0M/5e-4M	Gliadin concentration inside the cell	[30]
casein_aptazyme	0M	Concentration of the casein detecting aptazyme inside the cell	N/A
galactose_aptazyme	0M	Concentration of the galactose detecting aptazyme inside the cell	N/A
gliadin_aptazyme	0M	Concentration of the gliadin detecting aptazyme inside the cell	N/A
T7ptag	0M	Concentration of modified T7 RNA polymerase mRNA inside the cell	N/A
supD	0M	supD tRNA concentration inside the cell	N/A
eforRed	0M	eforRed chromoprotein concentration inside the cell	N/A
amilCP	0M	amilCP chromoprotein concentration inside the cell	N/A
casein_complex	0M	Concentration of the casein detecting aptazyme bound to casein	N/A
gliadin_complex	0M	Concentration of the gliadin detecting aptazyme bound to gliadin	N/A
galactose_complex	0M	Concentration of the galactose detection aptazyme bound to galactose	N/A
T7	0M	Concentration of the T7 RNA polymerase inside the cell	N/A
eforRed_RNA	0M	Concentration of eforRed mRNA inside the cell	N/A
amilCP_RNA	0M	Concentration of amilCP mRNA inside the cell	N/A

Table 7. Table of reactions used in outlined mathematical model, excluding degradation reactions

Name	Reaction	Rate	Description	Reference
casein_apt_trsc	null -> casein_aptazyme	k_trsc	Transcription of the casein aptazyme following mass action kinetics	[33]
galactose_apt_trsc	null -> galactose_aptazyme	k_trsc	Transcription of the galactose aptazyme following mass action kinetics	[33]
gliadin_apt_trsc	null -> gliadin_aptazyme	k_trsc	Transcription of the gliadin aptazyme following mass action kinetics	[33]
casein_complexation	casein + casein_aptazyme <-> casein_complex	k_on*casein*casein_aptazyme - k_off*casein_complex	Complexation of casein with casein aptazyme	[33]
galactose_complexation	galactose + galactose_aptazyme <-> galactose_complex	k_on*galactose*galactose_aptazyme - k_off*galactose_complex	Complexation of galactose with galactose aptazyme	[33]
gliadin_complexation	gliadin + gliadin_aptazyme <-> gliadin_complex	k_on*gliadin*gliadin_aptazyme - k_off*gliadin_complex	Complexation of gliadin with gliadin aptazyme	[33]
casein_cleave	casein_complex -> T7ptag	k_cleave*casein_complex	Cleavage of casein aptazyme	[33]
galactose_cleave	galactose_complex -> supD	k_cleave*galactose_complex	Cleavage of galactose aptazyme	[33]
gliadin_cleave	gliadin_complex -> amilCP_RNA	k_cleave*gliadin_complex	Cleavage of gliadin aptazyme	[33]
t7_trsc	T7ptag + supD -> T7ptag + supD + T7	k_trsl_T7*T7ptag*(K_T7*supD/(1+K_T7*supD))^2	SupD mediated transcription of T7 from T7ptag	[78]
eforred_trsc	T7 -> T7 + eforRed_RNA	k_trsc_T7*T7	Transcription of eforRed mRNA	[33]
eforred_trsl	eforRed_RNA -> eforRed + eforRed_RNA	k_trsl*eforRed_RNA	Translation of eforRed chromoprotein	[33]

Name	Reaction	Rate	Description	Reference
amilcp_trsl	amilCP_RNA -> amilCP + amilCP_RNA	k_trsl*amilCP_RNA	Translation of amilCP chromoprotein	[33]

Table 8. Table describing the expected results from each trial. All plates contain LB-agar, kanamycin, and chloramphenicol. The presence (+) or absence (-) of galactose, casein, and gliadin are indicated. Trials that do not contain any ligand or do not contain gliadin and both galactose and casein should not have coloured colonies. Trials that contain gliadin but do not contain both galactose and casein should have blue colonies from amilCP expression. The trial that contains both casein and galactose should have red colonies from eforRed expression, and the trial that contains all three ligands should have purple colonies from both amilCP and eforRed expression.

Galactose	Casein	Gliadin	Expected Outcome
-	-	-	Colourless
+	-	-	Colourless
-	+	-	Colourless
-	-	+	Blue
-	+	+	Blue
+	+	-	Red
+	-	+	Blue
+	+	+	Purple

Appendix B - Additional Information

Clarification on Aptazymes and Plasmid Assembly Methodology

The foundation of a responsive cell-based sensor lies in the integration of two distinct functional RNA domains: an aptamer and a hammerhead ribozyme [15]. Aptamers are synthetic, single-stranded nucleic acid sequences engineered through a process known as SELEX (Systematic Evolution of Ligands by Exponential Enrichment) to bind specific target molecules with high affinity and specificity [16]. While the aptamer serves as the detection unit, the hammerhead ribozyme acts as the catalytic engine, capable of undergoing self-cleavage at a specific phosphodiester bond [17]. By fusing these two components, we create an aptazyme; an allosterically regulated ribozyme whose catalytic rate is governed by the presence or absence of a target ligand [15].

The critical challenge in aptazyme engineering is ensuring that a binding event at the aptamer domain effectively communicates with the ribozyme core to trigger cleavage. This is achieved through a communication module, a short sequence of base pairs that physically connects the aptamer to one of the ribozyme's stems [15], [18]. This module acts as a structural bridge. Upon ligand binding, the aptamer undergoes a conformational shift that is transmitted through the communication module, either stabilizing or destabilizing the ribozyme's active site. Because the optimal sequence for this bridge is highly dependent on the thermodynamic stability of both the HHR and the specific aptamer used, the communication module is often identified or refined through in vitro selection (SELEX) or computational folding simulations to ensure high dynamic range in the sensor's response [15]. While SELEX is the gold standard for identifying these functional modules, it requires specialized high-throughput infrastructure. In the absence of such access, this experiment utilizes a validated communication module characterized by Yurdusev et al. 2025, which has demonstrated effective allosteric coupling within a similar structural framework [15].

To construct a functioning aptazyme, the hammerhead ribozyme, aptamer, and the signaling RNA molecule must be assembled in the correct order to ensure structural functionality. Following the methodology described in Yurdusev et al. 2025, the aptamer is inserted into the Stem III loop of the hammerhead ribozyme [15]. To establish the communication module, three-nucleotide oligonucleotides (GTA at the 5' end and TAC at the 3' end of the aptamer) are added to complete the bridge [15]. While mRNA is commonly fused directly to the 3' end of the aptazyme to allow for immediate translation upon cleavage, this architecture is modular and supports the inclusion of other RNA types [20]. Specifically, a target tRNA sequence can be integrated using the exact same fusion process as an mRNA sequence. This flexibility is essential for specialized components like the galactose aptazyme (which is designed to release a T7ptag tRNA), the catalytic mechanism remains identical regardless of whether the "payload" is a coding mRNA or a functional tRNA [20].

The proposed experiment will use three unique aptazyme-regulated expression cassettes; the DNA sequences encoding these aptazymes and their corresponding reporter genes will be cloned into a single plasmid vector and transformed into *E. coli*. Once transformed, the host machinery will constitutively transcribe these sequences into mRNA transcripts through constitutive promoters.

The coding sequence for the HHR sequence is as follows:

5'-GGACGAAAC| |GCGCTTCGGTGCGTCCTGGATTCCACGAAGGAGATATACC-3' [13].

The Stem III loop of the hammerhead ribozyme begins at the gap indicated by the vertical bars (| |), which serves as the precise insertion site for the communication module and the aptamer. The coding sequence for a gliadin aptamer is

5'-CTCTCGGGACGACCACTACGCATAGTTTCTATCGCCAGGAAGGGTCGTCCC-3' [19].

Adding the 3 base pair communication module consisting of a GTA linker at the 5' end and a TAC linker at the 3' end flanking the aptamer, inserting into the stem III loop of the HHR, and appending the DNA sequence corresponding to the mRNA sequence of interest to be translated to the 3' end of the sequence will yield a final functioning sequence:

5'-GGACGAAAC-GTA-CTCTCGGGACGACCACTACGCATAGTTTCTATCGCCAGGAAGGGTCGTC
CC-TAC-GCGCTTCGGTGCGTCCTGGATTCCACGAAGGAGATATACC-{mRNA}-3'

This sequence can then be positioned downstream of a constitutive promoter and upstream of a transcriptional terminator, ensuring a continuous production of the aptazyme.

Clarification on Appropriate Dosage

It has been demonstrated that 10% of patients with a milk allergy will react to amounts as low as 9.6mg of milk [79]. The density of cow milk is approximately 1.03g/mL, so 9.6mg of milk would be 9.32E-3mL in volume [80].

Casein, the milk allergen we test for, has a concentration in milk of 29.5g/L, or approximately 1.34mmol/L, given the mean molecular weight of caseins to 22kDa [81], [71]. Therefore, in 9.32E-3mL of milk, there would be about 1.25E-8mol of casein protein.

E. coli cells have a volume of 4.4E-15L on average [73]. Approximately 3.2E6 cells exist on a 60mm plate at confluency [73]. The total volume of all the cells would thus be 1.41E-8L.

If the milk and casein are assumed to disperse evenly, the concentration of casein in each cell would be equal to the average concentration, which can be calculated using the total amount of casein and the total volume. The total volume of the plate is the sum of cell volume and milk volume, coming out to be 9.33E-6L. The amount of casein was previously found to be 1.25E-8mol, so the average concentration comes out to be 1.34mM.

Performing similar calculations with galactose, which makes up 0.5g/L of milk and has a molar mass of 180.16g/mol, the concentration turns out to be 2.77mM [82], [83].

As for gluten, approximately 10mg can cause sickness in those with celiac [84]. Approximately 50% of gluten proteins are the gliadins tested for, with a mean molecular weight of 41.5kDa [85]. Again, performing calculations akin to those above yields an average concentration of 12.9mM.

It was decided to use 500µM as a ‘present’ signal, as it is below the concentrations calculated above, to test for sensitivity of the system so that it could respond to concentrations of allergens that can cause reactions in the vast majority of people. 500µM is also in line with other experiments run regarding protein concentrations in *E. coli,* confirming that this concentration of ligand is in the realm of possibility [30].

Minimum detectable concentration of amilCP and eforRed using Beer-Lampert law

The Beer-Lampert law states:

$$A = \epsilon \cdot l \cdot c \qquad (1)$$

Where A is absorbance (unitless), ϵ is the molar absorptivity constant with units of $mol^{-1}cm^{-1}$, l the distance light travelled through solution measured in cm, and c the concentration of the absorbing substance expressed in mol L^{-1}. For reliable detection, a minimum absorbance of $A_{Threshold} = 0.01$ is the standard threshold , and a path length of $l = 1\ cm$ is used for a standard cuvette [86], [87]. The molar absorptivity constant of amilCP and eforRed are $87,600$ and $31,100$ mol L^{-1} respectively [88]. Rearranging (1), the minimum detectable concentration $\left(c_{min}\right)$ can be found through:

$$c_{min} = \frac{A_{Threshold}}{\epsilon \cdot l} \qquad (2)$$

Substituting the respective values into (2), the minimum detectable concentrations of amilCP and eforRed are 0.11µM and 0.32µM respectively.

Appendix C - Supplementary Graphs

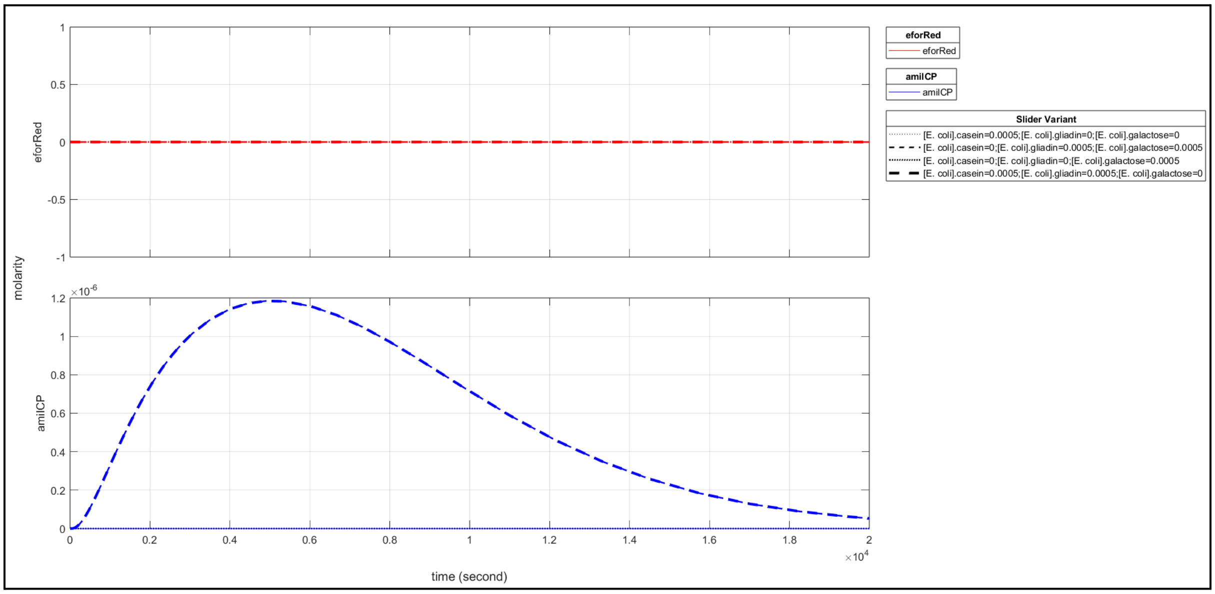

Figure 7. eforRed (red, top) and amilCP (blue, bottom) molarity over time shown in red, plotted under four different conditions: 500µM casein, 0M galactose, and 0M gliadin ( ), 0M casein, 500µM galactose, and 500µM gliadin ( ), 0M casein, 500µM galactose, and 0M gliadin ( ), and 0M casein, 0M galactose , and 500µM gliadin ( ). All other proposed test scenarios can be found in Figure 5.

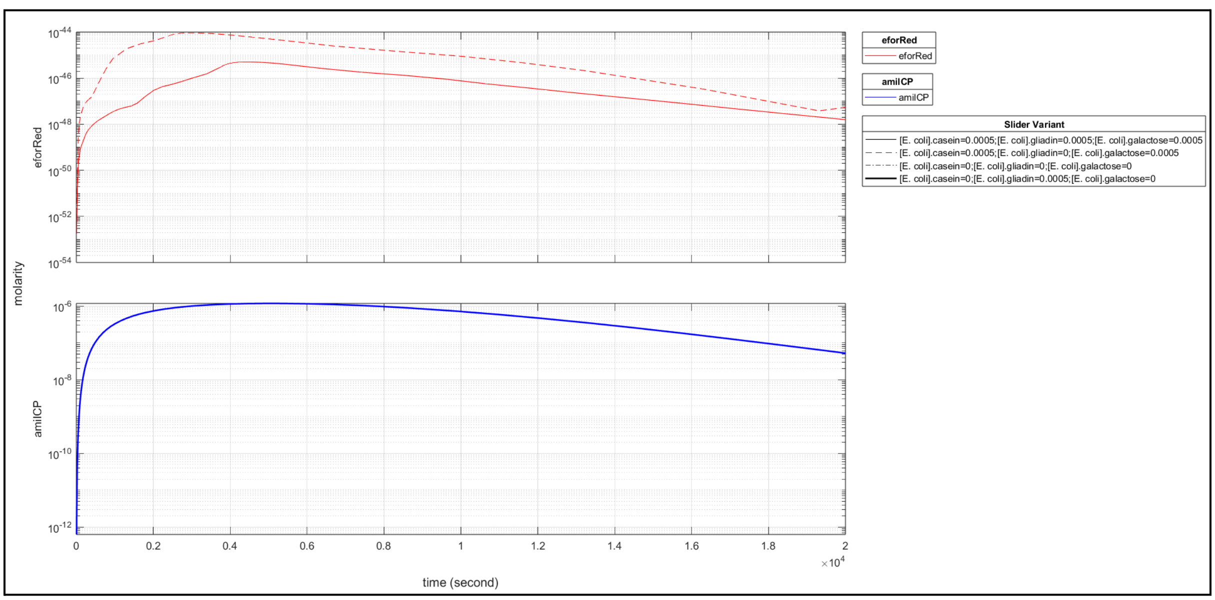

Figure 8. eforRed (red, top) and amilCP (blue, bottom) molarity over time shown in red, plotted on a semilog-y graph to better show parallel expression concentrations, under four different conditions: 500µM casein, 500µM galactose, and 500µM gliadin ( ), 500µM casein, 500µM galactose, and 0M gliadin ( ), 0M casein, 0M galactose, and 0M gliadin ( ), and 0M casein, 0M galactose , and 500µM gliadin ( ). All other proposed test scenarios appear as expected and have been omitted for clarity, but can be found in Figure 7 in Appendix C.